

Piyush Kumar Anand

Senior Software Engineer | Java | Spring Boot | Kafka | Reactive Systems | Cloud-Native Architecture

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PROFESSIONAL SUMMARY

Senior Software Engineer with 5.6+ years of experience designing and scaling cloud-native microservices across telecom, automotive, and compliance domains. Specialized in REST APIs with Feign Client + API Gateway (rate limiting, JWT, circuit breaker, fallback) and Reactive WebClient + API Gateway serving 50K+ concurrent users. Delivered measurable impact: 30% latency reduction, 20% throughput increase, 40% faster deployments. Skilled in Spring Boot, WebFlux, Reactor, RxJava, and Kubernetes deployments on AWS (EKS, EC2, Lambda). Recently expanded expertise into AI/LLM systems — integrating vector databases, embeddings, chunking strategies, and MCP protocol (Client/Server) for intelligent microservices. Recognized for mentoring teams, enabling scalable architecture adoption, and bridging enterprise backend engineering with AI-driven innovation.

TECHNICAL SKILLS

Languages: Java, JavaScript (ES6+), SQL, HTML/CSS

Frameworks: Spring Boot, Spring MVC, Spring WebFlux, Hibernate, React, Angular

Reactive Programming: Project Reactor, RxJava

Messaging & Integration: Kafka, RabbitMQ, REST APIs, JWT, Redis

Cloud & DevOps: AWS, GCP, Azure, Docker, Kubernetes, Jenkins, Git

Testing Tools: JUnit, Mockito, Jasmine, Jest, Cypress, Playwright

Databases: MySQL, Oracle, DB2

AI & LLMs: Large Language Models (LLMs), Vector Databases, Chunking & Embedding, MCP Client & Server

Architecture & Design: Microservices, API Gateway, Event-Driven Architecture, HLD/LLD

Others: Camunda, HashiCorp Vault, ELK Stack, Agile (Scrum)

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Capgemini

Bangalore | Dec 2024 - Jun 2025

- Architected Spring WebFlux microservices with WebClient + API Gateway, enabling real-time Kafka provisioning for 50K+ users.
- Implemented adaptive backpressure in Kafka pipelines, reducing latency by 30%.
- Automated CI/CD pipelines with Jenkins, Docker, and Kubernetes on AWS EKS, cutting deployment cycles by 40%.
- Designed Kubernetes manifests (Deployments, Services, ConfigMaps, Secrets) integrated with AWS workloads.
- Elevated Angular + Cypress test coverage to 90%+, improving release confidence.
- Mentored engineers on Project Reactor and reactive design patterns.

Design Engineer | British Telecom

Bangalore | Nov 2022 - Dec 2024

- Architected Spring MVC REST APIs with Feign Client + API Gateway (rate limiting, JWT, circuit breaker, fallback) for broadband provisioning, supporting millions of customer activations.
- Migrated APIs to Spring WebFlux, boosting scalability under heavy traffic.
- Optimized Kafka partitioning and consumer groups for high-throughput event delivery.
- Led OSB microservices migration in ESB project, consolidating middleware and improving interoperability.
- Introduced Playwright automation, reducing QA effort by 25%.

Senior Software Engineer | Renault Nissan Technology

Chennai | Nov 2019 - Nov 2022

- Migrated legacy Struts services into Spring Boot microservices, improving modularity and maintainability.
- Built RxJava-based compliance reporting pipelines, enabling real-time validation and regulatory adherence.
- Designed Kafka ingestion pipelines with schema registry and replay, improving data integrity and throughput by 20%.
- Increased JUnit + Mockito coverage to 95%, reducing production defects.

KEY PROJECTS

C Datacom (2024-2025)

- Developed Spring Boot + Kafka reactive microservices with exactly-once semantics, deployed on AWS EC2 + Lambda with Kubernetes.
- Integrated WebClient + API Gateway, achieving 30% latency reduction and 20% throughput increase.

ESB Integration (2022-2024)

- Designed REST APIs with Feign Client + API Gateway (rate limiting, JWT, circuit breaker, fallback) and Kafka + RabbitMQ reactive messaging to serve 50K+ users.
- Migrated OSB-based microservices into unified Spring Boot SOAP-based microservices.

BDTVV - Pollution Estimation (2019-2022)

- Migrated legacy Struts services to Spring Boot microservices, modernizing architecture and boosting maintainability.
- Designed Kafka pipelines with Redis caching, increasing telemetry throughput by 20%.

CCV Batch & Web (2019-2022)

- Reduced report generation time by 40% via Apache Beam pipelines on GCP Dataflow.

BDCO2 Tax Calculation (2019-2022)

- Automated CO2 emission tax calculations with Spring Boot + Oracle DB, improving reporting accuracy and compliance traceability.

EDUCATION

Master of Computer Applications (MCA) - IGNOU, New Delhi | Graduated: 2014

Bachelor of Computer Applications (BCA) - IGNOU, New Delhi | Graduated: 2011

CORE STRENGTHS

- REST APIs with Feign Client + API Gateway for secure, resilient integrations
- Reactive programming with WebClient, Reactor, RxJava, Kafka + RabbitMQ
- System design and scalability (HLD, LLD, fault tolerance)
- CI/CD automation with Jenkins, GitHub Actions, Docker, Kubernetes on AWS
- Code quality and automated testing (90-95%+ coverage)
- Agile and cross-functional collaboration
- Leadership and mentorship in modern reactive systems